1Z0-1157-26 Agentic AI Foundations Associate

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Oracle AI Database & Vector Search Foundations

Q1. Which approaches can generate embeddings for Oracle AI Vector Search workflows?

A. Export data to a Python environment, generate embeddings, then reimport via SQL*Loader

B. Train a custom embedding model and manually insert vectors row by row

C. Use external REST APIs exclusively, since in-database embedding is not supported

D. Use ONNX models in-database or external REST-based embedding providers ✓

Selected Answer: D **Analysis:** Oracle AI Vector Search allows users to leverage in-database embeddings directly using imported ONNX format models or by orchestrating network calls out to remote third-party REST embedding providers.

Q2. Why is chunking necessary before generating embeddings for large documents?

A. To convert text into SQL-compliant rows

B. To automatically encrypt content before indexing

C. To overcome token limits for large documents ✓

D. To remove semantic meaning from the document

Selected Answer: C **Analysis:** Foundational models and embedding generators have maximum context token limits. Large text payloads must be broken into optimal chunk segments to avoid payload truncation errors.

Q3. In the OpenAI Agents SDK, what is the role of the Runner?

A. It handles authentication and rotates API keys for the agent.

B. It is the deployment platform for hosting agents in production.

C. It executes the agent loop. ✓

D. It defines JSON schemas for function tools at compile time.

Selected Answer: C **Analysis:** The Runner is the operational engine within the SDK responsible for executing the primary loop, managing turn completions, and cycling interaction iterations between models and runtime environments.

Q4. How are tool calls handled between the LLM and the application?

A. It allows tools to run without application control.

B. It grants the orchestration layer unrestricted system access.

C. It generates a tool-call request for the application to validate and run. ✓

D. It performs the external operation directly during inference.

Selected Answer: C **Analysis:** Models emit an explicit structured tool call payload back to the client application, allowing the runtime engine to inspect, authorize, and run the physical utility code safely.

Q5. In JSON-RPC 2.0, what is the difference between a request and a notification?

A. A request includes an ID and expects a reply, while a notification does not. ✓

B. A request can carry structured data, while a notification can carry only text.

C. A request is sent unencrypted, while a notification is encrypted before processing.

D. A request uses HTTP, while a notification uses a different transport protocol.

Selected Answer: A **Analysis:** JSON-RPC specifications mandate that bidirectional standard requests contain a transactional correlation client identifier parameter ('id') and require response confirmations, whereas uni-directional notifications do not.

Q6. Which statement describes the STDIO transport in MCP?

A. STDIO transport requires OAuth authentication for all requests

B. The host launches the server as a local subprocess communicating via stdin and stdout ✓

C. STDIO transport serializes messages as plain text rather than JSON

D. The server runs as an HTTP service intended for remote multi-client deployments

Selected Answer: B **Analysis:** Under the Model Context Protocol (MCP), a local standard execution topology sets up the primary application host launching tool servers as nested subprocess units via stdin and stdout streams.

Q7. In OpenAI Agents SDK, how does the model select which tool to call?

A. It selects a tool randomly unless hardcoded.

B. It calls every registered tool before answering.

C. It selects the first registered tool by name.

D. It uses tool names, descriptions, and schemas. ✓

Selected Answer: D **Analysis:** The model calculates tool mapping matches contextually by analyzing structural properties exposed within the configuration array, specifically checking structural name bindings, descriptions, and input schemas.

Q8. What does the @function_tool decorator do in the OpenAI Agents SDK?

A. Caches the function's return value to disk for faster subsequent calls.

B. Automatically retries the function if it raises an exception during execution.

C. Registers the function as an HTTP endpoint the agent calls via REST.

D. Converts a regular Python function into a tool the agent can call. ✓

Selected Answer: D **Analysis:** Applying the SDK function tool annotation wraps regular programmatic routines into structured, model-callable functional entities complete with automatic schema inference mechanics.

Q9. Which native column type does Oracle AI Database use for storing vector embeddings?

A. VECTOR ✓

B. BLOB

C. JSON

D. VARCHAR2

Selected Answer: A **Analysis:** Oracle Database introduced the specific native `VECTOR` data abstraction structure to natively house array-based multi-dimensional numerical representations of mathematical data structures.

Q10. What is an embedding in a semantic search workflow?

A. A type of SQL JOIN designed for nested tables.

B. A database trigger that fires before INSERT.

C. A vector produced by a neural network. ✓

D. A compressed video file format used by streaming services.

Selected Answer: C **Analysis:** An embedding is a multi-dimensional array representation of dense continuous numerical coordinates emitted via deep transformer layers capturing latent contextual meanings of text blocks.

OCI Enterprise AI Infrastructure & Memory Paradigms

Q11. Which three model categories are available through OCI Enterprise AI Models?

A. Forecasting, Clustering, and Regression models.

B. SQL, NoSQL, and Graph models.

C. Chat models, Embed models, and Rerank models. ✓

D. Robotics, Audio, and Vision models.

Selected Answer: C **Analysis:** OCI Enterprise Generative AI services present discrete endpoints specifically scoped to text conversational interactions (Chat), textual geometric translations (Embed), and context verification operations (Rerank).

Q12. Which three layers make up the OCI Enterprise AI Platform?

A. AI Models, AI Agents, and AI Governance ✓

B. AI Models, AI Infrastructure, and AI Observability

C. Models, Containers, and Object Storage

D. Compute, Networking, and Storage services

Selected Answer: A **Analysis:** The architecture defines structural enterprise alignment through three main layers: foundational LLM models, orchestration interfaces (AI Agents), and strict compliance wrappers (AI Governance).

Q13. What is short-term memory compaction in OCI Enterprise AI Agents?

A. A network optimization for agent requests

B. A summarization process for long conversations ✓

C. A masking process for prompt content

D. A runtime optimization for Python tools

Selected Answer: B **Analysis:** To prevent exceeding hard max context bounds, OCI employs algorithmic context cleaning that compresses extensive message histories into consolidated relational summaries.

Q14. What is the purpose of the Vector Stores API?

A. Indexing and retrieving data by meaning. ✓

B. Translating text between languages.

C. Encrypting Object Storage buckets.

D. Streaming video into AI agents.

Selected Answer: A **Analysis:** The Vector Stores API orchestrates operations targeting indices optimized for calculating approximate semantic metrics, matching unstructured files according to contextual meaning.

Q15. What is long-term memory in OCI Enterprise AI Agents?

A. The model's pretraining corpus.

B. Durable memory shared across conversations. ✓

C. A static training dataset.

D. Persistent container block storage.

Selected Answer: B **Analysis:** Long-term memory models establish architectural data persistence frameworks that allow profile configurations and prior thread contexts to span across separate user session tracks.

Q16. Which set lists built-in tool categories supported by OCI Enterprise AI Agents?

A. SSH, FTP, and RDP tools.

B. SMS, Email, and Fax connectors.

C. VCN routing and load balancing tools.

D. File Search, Code Interpreter, Function Calling, and MCP Calling. ✓

Selected Answer: D **Analysis:** OCI orchestrates specialized processing boxes including integrated semantic document finders, programmatic code containers, downstream generic web functions, and MCP endpoints.

Q17. Which four behaviors does every Select AI Agent perform?

A. Retrieve, Rank, Generate, and Validate

B. Plan, Execute, Monitor, and Scale

C. Plan, Use tools, Reflect, and Remember ✓

D. Plan, Use tools, Summarize, and Terminate

Selected Answer: C **Analysis:** Modern task-executing setups utilize recursive loops matching four core processing states: task pathing (Plan), execution calls (Use tools), condition monitoring (Reflect), and data storage (Remember).

Q18. When is MCP most valuable?

A. For simple local utilities like add() and multiply()

B. When building a single-agent app with no external dependencies

C. When defining prompts and output parsers inside a single LangChain chain

D. When tools are external or cloud-hosted ✓

Selected Answer: D **Analysis:** MCP provides maximum deployment benefits when abstracting interfaces pointing to decoupled, complex, or cloud-hosted operational microservices, eliminating custom API integration layers.

Q19. In the context of MCP, what does the USB-C for AI analogy emphasize?

A. MCP needs specialized hardware to run.

B. MCP is preinstalled on all modern computers.

C. MCP is a standardized interface. ✓

D. MCP makes AI models physically faster.

Selected Answer: C **Analysis:** The primary USB-C comparison emphasizes defining a universal, structured protocol link standard that enables components to hook together out-of-the-box regardless of internal design.

Q20. Which prompt addition is used for zero-shot Chain-of-Thought prompting?

A. "Upload a structured CSV first."

B. "Set temperature equal to zero."

C. "Let's think step by step." ✓

D. "Disable all external retrieval tools."

Selected Answer: C **Analysis:** Appending phrases like 'Let's think step by step' alters the probabilistic output mapping of transformers, inducing structural multi-turn hidden calculations prior to emitting final strings.

Model Context Protocol (MCP) Standardized Architectures

Q21. Which MCP primitive is application-controlled and provides read-only context data to the model?

A. Resources ✓

B. Sessions

C. Tools

D. Prompts

Selected Answer: A **Analysis:** Resources behave as standard readable file schemas managed through the client application shell, injecting explicit static contextual text streams into active prompt histories.

Q22. Which statement describes an MCP Host?

A. It defines the protocol messages between clients and servers.

B. It executes tool implementations directly and returns results to the LLM.

C. Coordinates the LLM, user interaction, and MCP client connections to servers. ✓

D. It is the external REST API wrapped by the MCP server.

Selected Answer: C **Analysis:** The MCP Host behaves as the master consumer terminal system (e.g., orchestration application shell), managing model lifecycles, streaming user interfaces, and binding protocol clients.

Q23. In the OpenAI Agents SDK, when are input guardrails and output guardrails evaluated?

A. Input guardrails run before the agent processes input; output guardrails run before the response is returned. ✓

B. Input guardrails and output guardrails run only if explicitly triggered by the model.

C. Input guardrails and output guardrails both are evaluated only before the model receives user input.

D. Input guardrails and output guardrails both are evaluated only after the final response is generated.

Selected Answer: A **Analysis:** Input defensive filters intercept inbound string payloads before committing them to active model steps, while output compliance nodes filter generated tokens before rendering them to consumers.

Q24. Which statement describes an LLM-based AI agent?

A. A specialized transformer architecture.

B. A chatbot interface with additional visual components.

C. A newly trained standalone model type.

D. A model-based system that uses tools and orchestration. ✓

Selected Answer: D **Analysis:** An agent is a compound software layout where an underlying model acts as the core reasoning engine, utilizing structural routing paths and tool connectors to alter external file environments.

Q25. Which behavior is NOT a characteristic of modern LLM-based AI agents?

A. Pursuing goals across multiple steps

B. Requiring every execution path to be predefined ✓

C. Using external APIs or databases

D. Adjusting behavior based on observations

Selected Answer: B **Analysis:** Unlike deterministic workflows, agents exhibit non-linear dynamic behavior by deciding task execution paths recursively based on observation evaluations, instead of hardcoded static maps.

Q26. In the given LangChain chain, what is the role of StrOutputParser()? chain = prompt | model | StrOutputParser()

A. It sends the prompt to the model API.

B. It extracts plain text from the model response. ✓

C. It stores tool execution history.

D. It converts Python functions into tools.

Selected Answer: B **Analysis:** `StrOutputParser()` extracts the target textual message token components out of inbound base model chat container structures, streamlining down-pipe standard string parsing.

Q27. Which description defines memory poisoning in AI-agent systems?

A. Long prompts causing temporary context loss.

B. A SQL injection attack against a database.

C. Malicious content inserted into persistent memory stores. ✓

D. Hardware instability affecting RAM reliability.

Selected Answer: C **Analysis:** Memory poisoning constitutes an indirect injection attack vector where malicious strings find permanent placement within dynamic history states, corrupting subsequent inferences.

Q28. Why are docstrings especially important when defining LangChain tools with the @tool decorator?

A. They determine which API provider LangChain uses.

B. They encrypt tool arguments before execution.

C. They improve Python execution speed.

D. They tell the LLM when and how to use the tool. ✓

Selected Answer: D **Analysis:** LangChain extracts python method docstrings automatically to construct structural tool functional catalog logs which are injected directly into system prompts for model evaluation.

Q29. In the OpenAI Agents SDK, how does a Handoff differ from the Manager pattern?

A. A Handoff transfers control; a Manager keeps control. ✓

B. A Handoff uses async execution; a Manager uses sync execution.

C. A Handoff calls tools directly; a Manager only routes messages.

D. A Handoff keeps control; a Manager transfers control.

Selected Answer: A **Analysis:** A handoff passes execution control out to an independent execution target completely, while a coordinator loop pattern (Manager) retains the primary orchestration thread state locally.

Q30. What is Oracle AI Database Private Agent Factory?

A. A hardware appliance for training models on-premises

B. A no-code platform for building agents ✓

C. A backup utility for embeddings

D. A managed orchestration layer for deploying to Kubernetes

Selected Answer: B **Analysis:** The Private Agent Factory provides an integrated, security-isolated workspace within the Oracle converged framework to spin up conversational tool agents with zero code.

Orchestration Frameworks: LangChain & OpenAI Agents SDK

Q31. From the LLM's perspective, what is consistent between MCP-served tools and locally defined tools?

A. The LLM interacts with both through the same tool-calling interface. ✓

B. The LLM can only invoke MCP-served tools after explicit user approval.

C. MCP-served tools always return richer outputs than local tools.

D. The LLM receives network paths and authentication credentials for MCP tools.

Selected Answer: A **Analysis:** From the model's standpoint, tool parsing payloads share identical standard execution schemas; it remains unaware of whether the application calls a local function or a remote MCP node.

Q32. What is the strategic theme behind agentic AI capabilities in Oracle AI Database?

A. Bringing AI capabilities natively into the database ✓

B. Selling vector products without database integration

C. Removing SQL functionality in favor of REST APIs

D. Replacing the database with a vector-only system

Selected Answer: A **Analysis:** Oracle's primary goal centers on unifying compute mechanics by locating advanced search structures, tensor calculations, and model execution steps directly inside the secure data tier.

Q33. Which approaches are supported by OCI Enterprise AI Agents?

A. Select a free or paid deployment model

B. Call the Responses API directly or deploy a hosted agent application ✓

C. Build a frontend agent or a backend agent

D. Deploy using Python-only or Java-only options

Selected Answer: B **Analysis:** OCI structures enterprise interaction along two primary pipelines: directly binding programmatic workflows via standard API consumption or running a fully pre-configured UI instance.

Q34. Which statement describes the purpose of the OpenAI Responses API?

A. It compresses prompts to reduce the number of tokens billed.

B. It sends input to an OpenAI model and returns generated output. ✓

C. It hosts open-source models locally on the developer's own hardware.

D. It trains language models from scratch on customer-supplied data.

Selected Answer: B **Analysis:** The OpenAI Responses interface acts as a transactional client wrapper node whose fundamental task is passing user instruction schemas out to external models and catching generated target responses.

Q35. Which tasks is handled automatically by LangChain when using agent.invoke()?

A. Managing conversation state, formatting API requests, and routing tool execution across the agent loop. ✓

B. Optimizing GPU memory allocation and distributing model inference across hardware accelerators.

C. Training the foundation model from scratch.

D. Building tool schemas, parsing tool calls, and orchestrating execution loops.

Selected Answer: A **Analysis:** Calling `.invoke()` tells LangChain to boot its autonomous loop container, which updates session history keys, translates schemas into network request bundles, and guides tool selections.

Q36. Which SQL function computes distance between vectors in Oracle AI Vector Search?

A. VECTOR_DISTANCE() ✓

B. VECTOR_SCORE()

C. EMBEDDING_DISTANCE()

D. SCORE_SIMILARITY()

Selected Answer: A **Analysis:** The `VECTOR_DISTANCE()` method allows users to evaluate vector similarity directly using standard mathematical distance metrics (e.g., COSINE, EUCLIDEAN, DOT) in a SQL query.

Q37. What is the architectural advantage of Autonomous AI Database MCP Server over a separately deployed third-party MCP server?

A. It moves all query execution to a separate compute tier outside the database.

B. It eliminates the need for SQL queries entirely.

C. It reduces operational overhead and enforces security inside the database. ✓

D. It uses a common MCP host for all database connections.

Selected Answer: C **Analysis:** Embedding the protocol components within the Autonomous pipeline eliminates additional network connection layers and aligns security controls with native row-level protections.

Q38. Which OCI services are used for observability and auditing of deployed AI agents?

A. OCI Streaming, OCI Queue, and OCI Email Delivery

B. OCI Logging, OCI Monitoring, and OCI Audit ✓

C. OCI Events, OCI Functions, and OCI Vault

D. OCI Data Catalog, OCI Search, and OCI DNS

Selected Answer: B **Analysis:** OCI's management platform tracks service logs through OCI Logging, exposes execution telemetry through OCI Monitoring, and records configuration modifications via OCI Audit.

Q39. Which authentication approach should be used for production-grade access to OCI Enterprise AI services?

A. OCI IAM authentication with signed requests and IAM policies ✓

B. Hardcoded credentials checked into source repositories

C. Anonymous access to tenancy-level resources

D. Browser session cookies stored with application code

Selected Answer: A **Analysis:** Production deployment pipelines require OCI Identity and Access Management (IAM) configurations, validating cryptographic requests against resource access policies.

Q40. Which Python package is installed first for a simple OCI Responses API setup?

A. openai ✓

B. requests-html

C. boto3

D. django

Selected Answer: A **Analysis:** The OCI Enterprise AI Responses architecture integrates with industry-standard protocol designs, utilizing the base `openai` distribution wrapper client to talk to its unified model gateway endpoints.